

Chih-Wei Joshua Liu

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EDUCATION

Leland Stanford Junior University Biophysics PhD Program · Thesis advisor: Grant M. Rotskoff	Stanford, CA, USA 09/2024 – present
Massachusetts Institute of Technology (MIT) Bachelor of Science in Physics – Focus Area in Chemistry GPA: 4.9/5.0 · MCAT: 528/528	Cambridge, MA, USA 05/2022
National Experimental High School SAT: 1600/1600 · ACT: 36/36	Hsinchu, Taiwan, ROC 05/2018

SELECTED PROJECTS

C.-W.J. Liu, J. Klinger, G.M. Rotskoff, Optimal parameterization of nonequilibrium generalized master equations from discrete-time experimental data, [arXiv:2606.28289](https://arxiv.org/abs/2606.28289), in revision at *Phys. Rev. X* (2026). [\[code\]](#)

- Derived the first maximum-likelihood parameterization of non-Markovian master equations validated on real experimental data and satisfying all physical constraints in general stochastic systems.
- Set new state-of-the-art among coarse-grained models in reproducing experimentally measured folding and relaxation rates of benchmark HP35-Lys24Nle/Asn27His/Lys29Nle protein from atomistic molecular-dynamics simulation.

J. Li*, **C.-W.J. Liu***, M. Szurek, and N. Fakhri, **Measuring irreversibility from learned representations of biological patterns**, *PRX Life* **2**, 033013 (2024), *equal contribution. [\[code\]](#)

- Extended thermodynamic-irreversibility estimation to entire experimental videos of high-dimensional biological processes using variational-autoencoder representations.

H. Lee*, E. Koskelo*, S. Gokhale*, J. Li*, C. Fei, **C.-W.J. Liu**, L. Lin, J. Dunkel, D.J. Skinner, and N. Fakhri, **Topological flowscape reveals state transitions in nonreciprocal living matter**, *Phys. Rev. X* **16**, 031061 (2026), *equal contribution.

- Trained a classifier to distinguish particle classes with nonreciprocal interactions, enabling analyses of previously experimentally unobserved nonequilibrium states.
- Featured in *Physics* magazine and on the cover of *Phys. Rev. X*.

Change-point detection with guaranteed convergence · [py-step-detect](#)

Developed `scikit-learn`-style Python interface for detecting change points in noisy time series, implementing the current state-of-the-art dynamic-programming-based algorithm.

POSITIONS

MIT Physics of Living Systems Group	Research Assistant (06/2022 – 08/2024)
MIT Department of Physics: Teaching Assistant, 8.241 Introduction to Biological Physics (09/2023 – 05/2024); Tutor, 8.044 Statistical Physics I (02/2021 – 05/2021); Teaching Assistant, 8.01 Physics I (09/2019 – 12/2019).	
Massachusetts General Hospital	Data Analysis Intern (06/2021 – 08/2021)

Analyzed patient statistics and authored two clinical publications.

LEADERSHIP AND HONORS

National Science Foundation Graduate Research Fellowship (2024, Physics).
National Defense Science and Engineering Graduate Fellowship (2024, Physics; declined for GRFP).
Sigma Pi Sigma: Physics honor-society inductee and MIT Chapter President (2022); **MIT News:** [profile \(2022\)](#).
USA Biology Olympiad: Top-50 High Honorist (2017); Semifinalist (2015–2018).
MIT Society of Physics Students: President (2021–2022).
Phi Kappa Theta – Massachusetts Eta: Service Chair (2019–2020); Pledgemaster (2021). Sourced and fundraised donation of 12,000 surgical masks to Beth Israel Deaconess Medical Center during pandemic shortage.

SKILLS AND INTERESTS

Computing: lightning, jax, pydantic, numpy, scipy, torch, scikit-learn, statsmodels, uv, ty, ruff, modal
Natural language: Standard Chinese (native)
Research interests: nonequilibrium statistical mechanics, time series, optimal transport, LLM expressivity
Personal interests: phonology, historical linguistics, international-relations theory, Midwest emo, ichthyology